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**Broselow**

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(54) **QUICK INFORMATION PORTAL**

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**G06K 7/14** (2006.01)

(52) **U.S. Cl.**  
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(2013.01)

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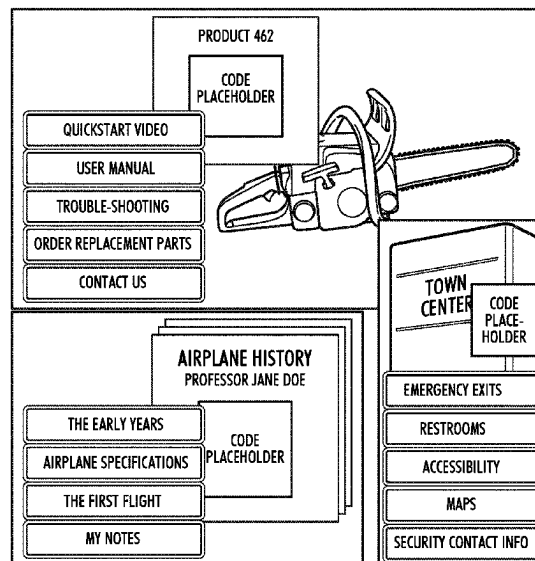
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Maynard Nexsen PC

(57) **ABSTRACT**

In one aspect, a computer implemented method for provid-  
ing rapid access to product information on a mobile com-  
puting device is disclosed. The method includes provision-  
ing an online platform and associating a database repository.  
Next, the method generates a visual code and identifies the  
visual code to a domain name. Then the visual code is  
populated with information, such as product information,  
wherein the product information is associated to the visual  
code that is further linked to the data repository and online  
platform. The association allows for multiple file types and  
acquisition from web scraping. Further, the online platform  
allows for editing content behind the domains and generat-  
ing further subdomains that further link to the original visual  
code. Thereby providing, in one aspect, a quick information  
portal through the use of visual codes and data structuring.

**20 Claims, 14 Drawing Sheets**



**Related U.S. Application Data**

(60) Provisional application No. 63/272,602, filed on Oct. 27, 2021.

(58) **Field of Classification Search**

USPC ..... 235/451

See application file for complete search history.

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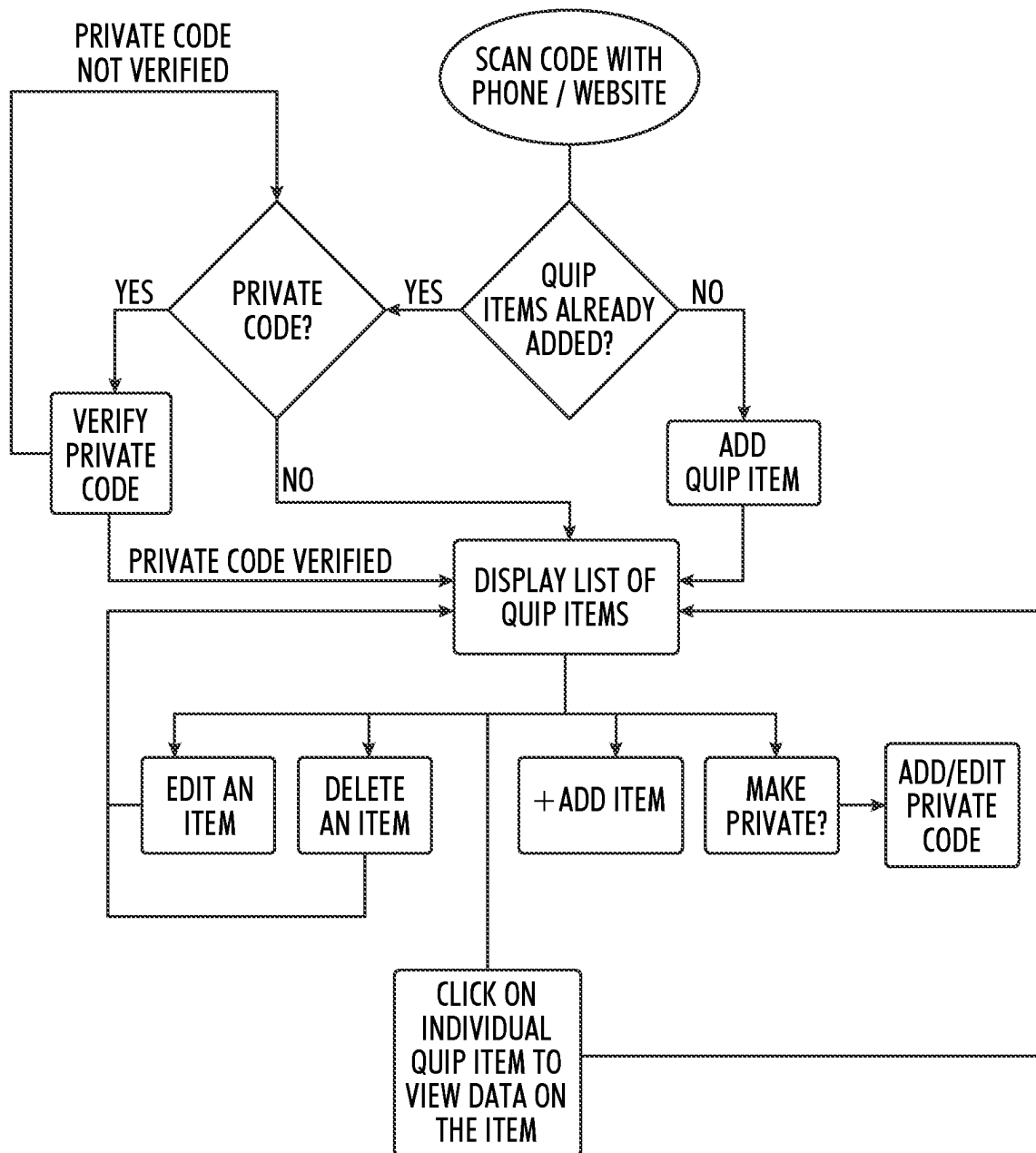
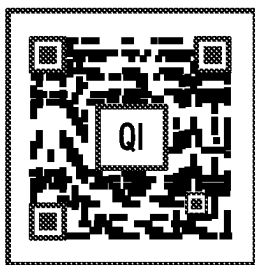


FIG.1



aA
 
 QI-CODE.COM

HOW DO YOU WANT TO ACCOMMODATE?:

TEXT

SUBMIT

^
v

DONE

TEXT

PHOTO

VIDEO

AUDIO

FIG.2

| EXAMPLE MYQUIP FOR A<br>VACATION RENTAL | CONTENT EXAMPLES                                     |
|-----------------------------------------|------------------------------------------------------|
| WELCOME MESSAGE                         | VIDEO WELCOME MESSAGE                                |
| HOUSE RULES                             | DOCUMENT                                             |
| WIFI AND APPLIANCE DETAILS              | DOCUMENT                                             |
| THINGS TO SEE AND DO                    | TEXT INCLUDING HYPERLINK TO TOWN INFORMATION WEBSITE |
| EMERGENCY INFO                          | DOCUMENT                                             |

FIG.3

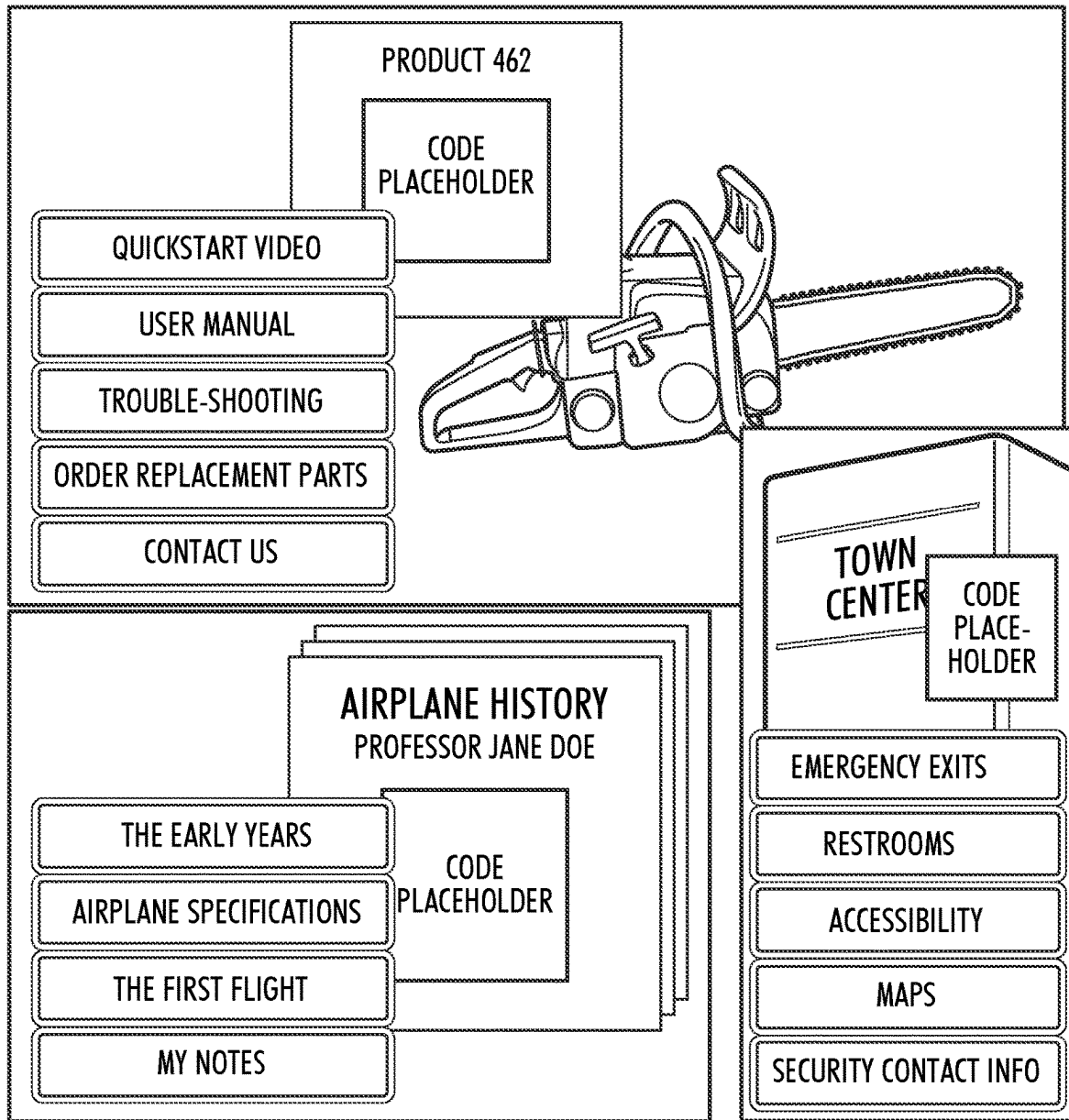


FIG.4

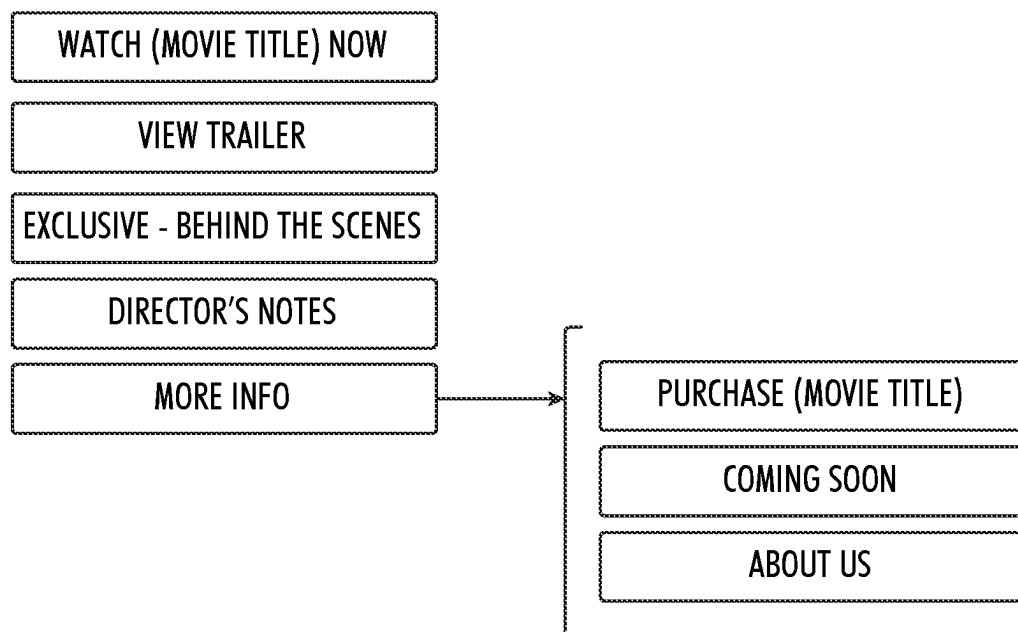


FIG.5

**CODE GENERATOR FORM**

TYPE OF DURATION:

NUMBER OF CODE:

ZIP NAME:

PRIVACY ACCESS CODE:

QI OPTIONS

QR CODE BORDER:

QR CODE LOGO:

EDITABILITY:

STORAGE DURATION (YEARS):

FIG.6



TYPE OF DURATION:

NUMBER OF CODE:

ZIP NAME:

PRIVACY ACCESS CODE:

QI OPTIONS

QR CODE BORDER:

☒ INCLUDE IMAGE FEATURE

QR CODE LOGO:

EDITABILITY:

STORAGE DURATION (YEARS):

☐ INCLUDE IMAGE FEATURE  
☐ ENABLED HOSPITAL NOTIFICATION

FIG.7

5 NUMBER OF BROSCODE GENERATED AFTER GENERATED QR CODES,  
WE WILL SEND YOU AN E-MAIL

TYPE OF DURATION:

PROQUIP ▼

NUMBER OF CODE:

5

ZIP NAME:

ROBIN TEST

PRIVACY ACCESS CODE:

PRIVACY ACCESS CODE

QI OPTIONS

QR CODE BORDER:

QI-BLUE ▼

☒ INCLUDE IMAGE FEATURE

QR CODE LOGO:

QI-BLUE ▼

EDITABILITY:

OPEN EDIT ▼

STORAGE DURATION (YEARS):

5

☒ INCLUDE IMAGE FEATURE

☐ ENABLED HOSPITAL NOTIFICATION

SUBMIT [DOWNLOAD CSV](#)

FIG.8

ITEM DESCRIPTION:

HOW DO YOU WANT TO COMMUNICATE?:

TEXT

FILE EDIT VIEW INSERT FORMAT TOOLS TABLE

↩ ↪ 🖼️ <>

MAKE CODE PRIVATE? ☐

LANGUAGE:

ENGLISH

TAB COLOR:

TEXT COLOR:

SUBMIT

FIG.9

HOW DO YOU WANT TO COMMUNICATE?

TEXT

SUBMIT

FIG. 10

HOW DO YOU WANT TO COMMUNICATE?

TEXT

TEXT

PHOTO

VIDEO

AUDIO

SUBMIT

FIG. 11

< BACK

OPERATOR MANUAL

DELETE 

DOCUMENT 

CHOOSE FILE

NO FILE CHOSEN

UPDATE

LINDEK10K13OPERATORMANUAL.PDF

TEXT 

DOCUMENT 

FILE EDIT VIEW INSERT FORMAT TOOLS TABLE

I

POWERED BY TINY

ADD

TEXT 

FILE EDIT VIEW INSERT FORMAT TOOLS TABLE

I

POWERED BY TINY

ADD

FIG.12

EDIT/DELETE ITEMS

OPERATOR MANUAL

SAFETY VIDEO

ENGLISH

+ ADD ITEM

MAKE CODE PRIVATE?

↕ REORDER?

ITEM DESCRIPTION:

PRODUCT PHOTO

HAS SUBQUIPS?: ☐

HOW DO YOU WANT TO COMMUNICATE?:

PHOTO

CHOOSE FILE

NO FILE CHOSEN

LANGUAGE:

ENGLISH

TAB COLOR:

TEXT COLOR:

SUBMIT

FIG.13

EDIT/DELETE ITEMS

OPERATOR MANUAL

SAFETY VIDEO

PRODUCT PHOTO

ENGLISH

+ ADD ITEM   MAKE CODE PRIVATE?    REORDER?

FIG.14

EDIT/DELETE ITEMS

OPERATOR MANUAL

SAFETY VIDEO

PRODUCT PHOTO

ENGLISH

+ ADD ITEM   MAKE CODE PRIVATE?    REORDER?

FIG.15

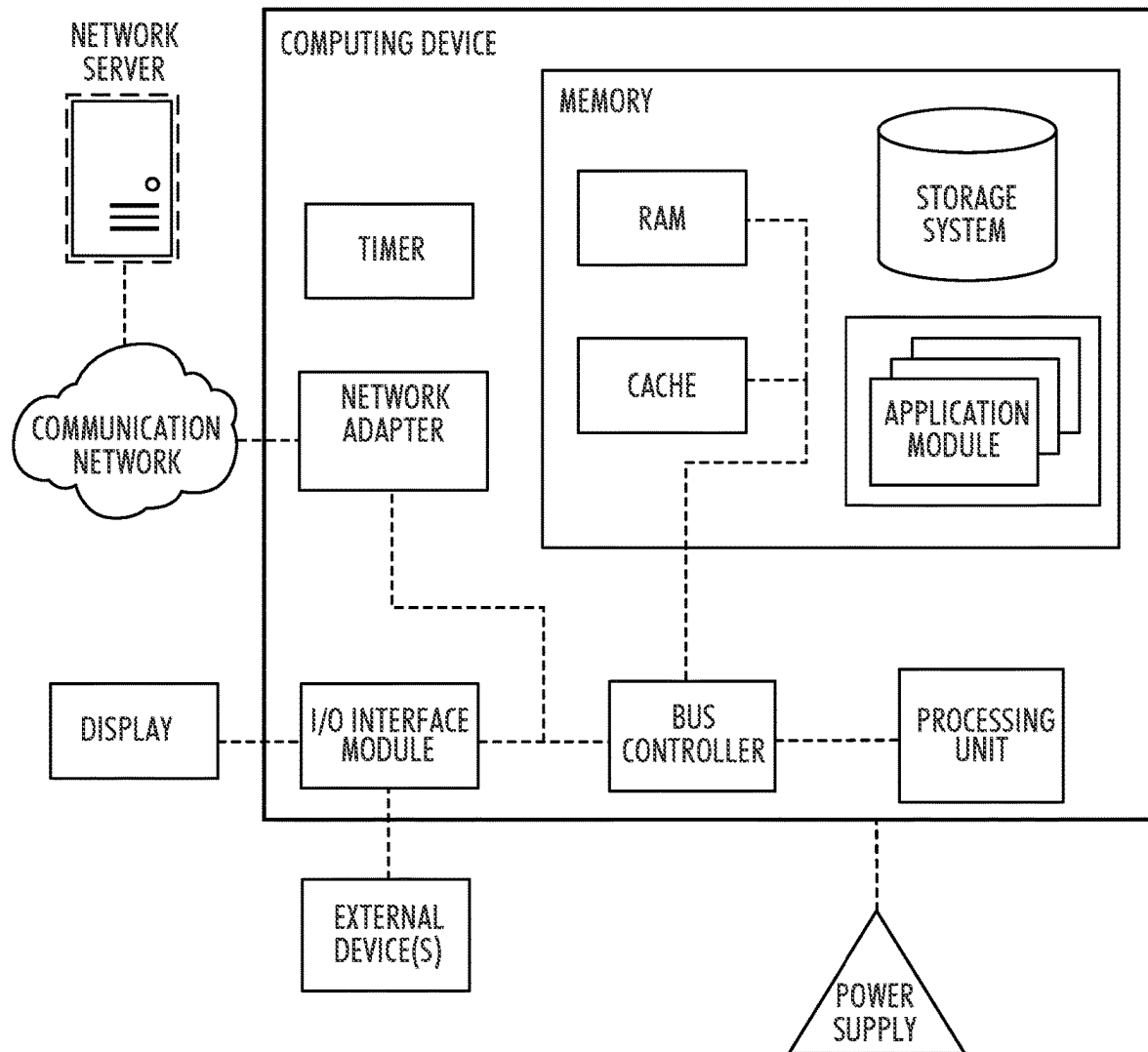


FIG.16



1

**QUICK INFORMATION PORTAL****CROSS-REFERENCE INFORMATION**

This application is a continuation-in-part of U.S. patent application Ser. No. 17/729,783 filed Apr. 26, 2022, which claims the benefit and priority of U.S. Provisional Patent Application No. 63/272,602, filed on Oct. 27, 2021, and titled Quick Information Portal (QUIP), the contents of each of which are herein incorporated by reference in their entirety.

**FIELD**

The present disclosure relates to computer implemented systems and methods for providing information access. In particular, computing systems and methods for gathering, formatting, curating, and distributing product and other information for access through a visual code and a mobile computing device.

**BACKGROUND**

Access to information such as product manuals, user guides, tutorials, video guides, and others are often found on the manufacturers website, if at all, and will typically require a deep understanding of the product to locate. For example, medical devices may have numerous models and editions, some even designated for country of origin and use, along with other derivative models such as for power requirements or otherwise. In doing so there is a dizzying array of choices and it becomes arduous and time consuming to locate material.

Further, information in general, when it relates to a specific category or topic, is difficult to maintain and often causes painstaking rewriting of html and css in order to present it digitally.

There remains a need to place a visual code on products or as part of a handout (sticker, mailer, information card) that curates the information as well as allows for rapid deployment and frequent low cost updates. The disclosure herein addresses this long sought need with a practical and efficient solution.

**SUMMARY**

This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used in isolation as an aid in determining the scope of the claimed subject matter.

Various aspects of computer and hardware implemented systems and methods are disclosed herein. In one aspect, a method is disclosed for providing rapid access to product information on a mobile computing device, such as a mobile phone or tablet. First, the method provisions an online platform located at a domain name and a database repository supporting the online platform. Next, the method includes generating, by the online platform at least one visual code that is scannable by the mobile computing device, wherein generating creates a subdomain and provisions a database location within the data repository for each visual code. Next, gathering product information and associating it with the at least one visual code directing to the subdomain. Then, formatting the product information based on at least the type

2

of content, wherein the formatting of the product information identifies it as a photo file, a video file, a pdf file, a sound file, or a text file and places a text label associated with a given file and further associates the given file with the at least one visual code. Next, storing the product information on the database repository of the online platform. Lastly, accessing, by the mobile computing device, the stored product information by scanning the at least one visual code.

In other aspects, a computer implemented method for providing instant access to information on a mobile computing device configured with a camera module is disclosed. The method first provisions an online platform located at a domain name and a database repository supporting the online platform. Next, generating by the online platform at least one visual code that is scannable by the mobile computing device, wherein generating creates a subdomain and provisions a database location within the data repository for each visual code. Then issuing, by the online platform, an image file containing the at least one visual code. Next, gathering information and associating it with the at least one visual code directing to the subdomain. Then, formatting the information based on at least the type of content, wherein the formatting of the information identifies it as a photo file, a video file, a pdf file, a sound file, or a text file and places a text label associated with a given file and further associates the given file with the at least one visual code. Then, storing the information on the database repository of the online platform. Then, scanning the at least one visual code by the mobile computing device configured with the camera module. Then, accessing the stored information by the mobile computing device. Lastly, editing the stored information from the mobile computing device accessing the stored information within the database repository on the online platform.

These and other embodiments are described in greater detail in the description which follows.

**BRIEF DESCRIPTION OF THE DRAWINGS**

Many aspects of the present disclosure will be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, with emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views. It should be recognized that these implementations and embodiments are merely illustrative of the principles of the present disclosure. In the drawings:

FIG. 1 is a flow chart illustrating an example method in the generating and or provisioning of and editing thereof of information related to a visual code;

FIG. 2 is an illustration of an example user interface wherein generating a visual code is processed;

FIG. 3 is an illustration of an example user interface viewing a hyperlink associated with a visual code;

FIG. 4 is an illustration of an example user interface viewing a hyperlink with information stored in a data repository associated with a visual code;

FIG. 5 illustrates an example user interface viewing subdomain information stored on a data repository associated with a visual code;

FIGS. 6-11 illustrates examples of a user interface of a visual code generator;

FIGS. 12-15 illustrates examples of a user interface for editing the information created and associated with a visual code;

FIG. 16 is a block diagram of an example computing device, such as a mobile computing device.

#### DETAILED DESCRIPTION

Implementations and embodiments described herein can be understood more readily by reference to the following detailed description, drawings, and examples. Elements, apparatus, and methods described herein, however, are not limited to the specific implementations presented in the detailed description, drawings, and examples. It should be recognized that these implementations are merely illustrative of the principles of the present disclosure and the invention. Numerous modifications and adaptations will be readily apparent to those of skill in the art without departing from the scope of the present technology.

#### I. Background Technology

Organizing information in useful ways is a continual process. Many inventors have progressed the science of storing information. Early on the library catalog system served as a useful means for organizing locations of texts. The internet and connected computers through the world wide web introduced an abundance of data. Along with that abundant data came problems with how to organize the information. Thus, searching algorithms were devised, and crawlers, along with meta tags and the like to assist in compiling data based off a query.

When QR codes were introduced they allowed a mobile computing device or camera to scan the code to retrieve data. That data is limited to a particular address and is often not editable. The disclosure herein advances techniques of QR codes to allow for accumulating information, such as product information, health information and more and building a hierarchy that allows for accessibility and direct editing by the owner of the visual code.

#### II. Example Embodiments

In one aspect, a computer implemented method for providing rapid access to product information on a mobile computing device is disclosed. A computer may be a server, or a standard general purpose computer as identified in FIG. 16. Further, the rapid access is provided through a visual code accessing an online platform. In the example method an online platform is provisioned and hosted on a server at a URL and associated with a domain name. In this example the visual code is not associated with any product or knowledge base on creation, thus generating a code prior to affixing it to a product, etc. In this aspect, manufacturers may provision codes and begin working consecutively on content even without the manufacturer. For example, social media insiders may be provisioned codes in which they can provide useful content, the manufacturer then may choose whether to activate the code on the product. Continuing, the provisioned online platform is configured with a database such as SQL database or other relational database that allows configuration and storage of information. In the example method the online platform services may be provisioned through a cloud server or hosting environment with built in libraries such as Amazon Web Services (AWS)<sup>TM</sup> or Microsoft Azure<sup>TM</sup> or Google Cloud<sup>TM</sup>.

In other aspects the online platform may be a web platform through a content management system such as WordPress<sup>TM</sup> that allows pairing with a database repository such as MySQL<sup>TM</sup> or MariaDB<sup>TM</sup> or other relational data-

bases. In either embodiment the core functionality remains the same, the online platform provides a form based generator that allows various inputs to generate a visual code. Further, there may be a host of open source plugins or content and libraries paired therewith. In the example, the generated visual code is scannable, and may resemble a QR code that includes border options and the placement of logos or other details in which to customize the scannable visual code. In one aspect, a mobile computing device with a camera is capable of scanning the visual code and then opening a web browser on the mobile computing device that points to a specific domain, wherein the domain is supported by a database repository.

The database repository may be a relational database repository and may be used to store the various information types, such as a photo file, a video file, a pdf file, a sound file, or a text file. Further, from an editing menu on the online platform a user may edit the information that is found at the data repository and such information is reflect to the same visual code. Additionally, a subdomain may be propagated or created and configured to the visual code, thus allowing generation of subdomains from a domain based on at least one visual code. Therefore, in such an arrangement, new information may be compiled and placed at disparate subdomains that all originate from one visual code and domain and that are all accessible from the one visual code.

In other aspects of the method, from the edit menu or creation menu, a formatting options allows editing or creating of the content for the various file types. In this example the edits are stored and pushed to the database repository allowing for rapid updates to information behind the visual codes, that is information stored in the data repository and assigned to a web domain. Further, the information is accessible by any means that allow for scanning of visual codes and opening and translating through a web browser. Thus mobile computing devices or computing devices with a camera are capable of accessing the information rapidly and without user input.

In additional embodiments the methods disclosed herein further comprise displaying the product information in a formatted version that is adapted to the mobile computing device to allow for accessibility and viewing. In this example the video files or other file types may be formatted to fit a particular screen. For example, a tablet computer possesses a larger dimension screen, and scaling of the content is provided through the online platform. In other embodiments scaling may be rendered on the device.

In another aspect, the steps may include creating additional subdomains for each domain linked to a data repository, in this aspect the additional subdomains may originate from the original visual code. Further, the content hosted on the data repository may be accumulated from data scraping (also known as data extraction or web scraping) of target domains, thus automating the intake or gathering of information. An example of a python library that may scrape the data is BeautifulSoup, which allows for rapid data acquisition and implements an ability to incorporate the data onto the data repository. There are many libraries that may assist with the acquisition of structured web data that may allow for automated gathering of information within a data repository linked to at least one visual code. Further, the data portion may be performed by third parties that are given access, thus allowing independent third parties to generate content that is associated to a visual code.

In one aspect, with respect to the visual code, at the generating step the visual code may also be provisioned with a duration code, a privacy access code, a border for styling

5

such as a blue border or a border to match an organizations preferred colors, a logo, and a setting for allowing edits, wherein edits may be placed on already issued visual codes by altering the data repository that is ultimately directed to a domain name. The ability to edit allows for not only editing post the generation of the visual code, but also to rapidly deploy codes and frequently update the information behind the code. In one aspect a visual code may serve as a digital journal, wherein updates are accessed by scanning the code and editing daily text. Similarly, the access code or privacy code allows only users with the code to access the information or to edit the information. When selecting the privacy access code a user may provide secured access for health information or personal identifiable information. In this aspect the visual code is protected and only accessible from users with a passcode. In other aspects the access code may be used for granting access to editing the content, but not restrict access. In the two differing embodiments the privacy access code may be used for editing or may be used for accessing in general, thereby allowing only approved users to edit content or approved users to view content.

The present technology may be embodied as, among other things, a system, method, or computer-product. Accordingly, embodiments may take the form of a hardware embodiment, a software embodiment, or an embodiment combining software and hardware. In one embodiment, the present invention takes the form of a computer program product that includes computer usable instructions embodied on one or more computer readable media and executed by one or more processors.

According to some embodiments, systems and methods for providing rapid access to information and/or content on a mobile computing device, such as a mobile phone or smartphone, or tablet are provided. In some aspects, systems and methods described herein can additionally provide for rapid editing or changing or otherwise modifying information and/or content that can be accessed by a computing device and/or mobile computing device.

In one aspect, an online platform can be provisioned and/or provided, the online platform being located at a domain or domain name, or URL. In some aspects the online platform can be implemented as an information portal platform or information portal engine. The information portal platform can be provisioned and hosted on a server at a URL and be associated with a domain or domain name, referred to herein as a main domain. In some aspects, the information portal platform or online platform can be configured or in communication with one or more databases such that the information portal platform is supported by one or more database repositories. The one or more database repositories can contain a plurality of resources, for instance digital resources. A database repository may be a relational database repository and may be used to store the various information types, such as a photo file, a video file, a pdf file, a sound file, a text file, and/or hyperlinks or URL information.

In some embodiments, a system or method for providing rapid access to information and/or content is configured to generate one or more information portals and/or sub-information portals, and at least one visual code. In some aspects, a visual code can be generated and associated with or otherwise linked to an information portal, and the information portal can further be associated with and/or linked to one or more generated sub information portals. In some aspects, the generation of a visual code, the information portal, and optionally the one or more sub information portals can further include the generation and/or propagation

6

of one or more subdomains, where each of the one or more subdomains is associated with the information portal and the one or more sub information portals. As will be appreciated, each subdomain can be propagated from or originate from the domain or main domain. In some aspects, the information portal can be associated with a first subdomain and the one or more sub information portals can be associated with one or more additional subdomains.

In some aspects, an information portal and/or one or more sub-information portals can incorporate one or more links or hyperlinks to one or more resources, such as a photo file, a video file, a pdf file, a sound file, a text file, and/or hyperlinks or URL information. An information portal and/or optionally one or more sub information portals can incorporate any combination of resources or resource information, that is, can be configured to provide a plurality of content destinations that stem from the first subdomain and/or additional subdomains. Additionally, in some embodiments, content destinations can be modified and/or deleted in the information portal and the optional one or more sub information portals without changing or redirecting any of the generated subdomains.

According to some embodiments, systems described herein can include an online platform, such as an information portal platform for the generation of a visual code, an information portal, and optionally, one or more sub information portals. The online platform can include an information portal generation component that provides a form based generator that allows various inputs to generate a visual code and the information portal and/or sub information portals. Accordingly, systems described herein can also include a visual code generation component, a linking component, and an information portal management component, among others. Based on one or more user inputs an information portal generation component can generate an information portal and in some instances one or more sub information portals, where the sub information portals can have any hierarchical configuration from the information portal. A visual code generation component can generate a visual code that can be associated with or correspond to the generated information portal. The linking component can link information to the information portal and/or one or more sub information portals, for example based on one or more user inputs or automatically using a crawler tool that returns relevant content. According to some aspects, an information portal management component can be configured to enable a user, such as an end user, to add, modify, view, and/or delete content and/or links in a generated information portal or sub information portal. The content and/or links can be added, modified, and/or deleted without changing or redirecting any of the underlying URL information or structure.

In some embodiments, systems and methods herein can generate an information portal template and associated visual code. In some aspects a form based generator can allow for inputs to generate an information portal (and optionally one or more sub information portals) templates, and a corresponding visual code, that can incorporate one or more placeholders in a standardized format for multimedia content and/or web hyperlinks to be subsequently added or linked to the information portal or a sub information portal. An information portal template and visual code can be utilized by an end user to then add, modify, view, and/or delete links or hyperlinks. In some aspects, an information portal described herein or information portal template can incorporate security features such as password protection functionality for any of the end user functions (e.g. add,

modify, delete). As will be appreciated, an information portal can incorporate a single master password or different passwords for different functions or end user roles.

In some embodiments, the online platform or information portal platform is configured to generate the visual code and information portal (and one or more sub information portals and associated hierarchy), and one or more subdomains simultaneously. According to some aspects, the information portal can be associated with a first subdomain, and one or more optional sub information portals can be associated with one or more other subdomains (e.g. second subdomain, third subdomain, nth subdomain). According to some aspects, hyperlinks located on, located on, or associated with the information portal can direct to the one or more sub information portals (e.g. a first level of sub information portals, a second level of sub information portals, nth level of sub information portals). The hierarchical information corresponding to an informational portal (and e.g. sub information portals, hierarchy information, associated and/or linked content or information), can be stored on a database or repository associated with the online platform (e.g. information portal platform) along with the visual code associated with the information portal. As will be appreciated, according to some aspects, an information portal can refer to all the above-mentioned aspects, including configuration and hierarchy, and incorporated data, datasets, or other information or content.

According to some embodiments, systems and methods are provided to generate on demand, template, shell, or just-in-time information portals. As will be appreciated in some aspects, an information portal may refer to the information portal and the associated visual code as a single concept, that is, when an information portal is built and generated, a visual code is also generated for that information portal. In some aspects, one or more inputs (e.g. by batch upload file) or user inputs can be input into a generator or form-based generator provided by the online platform. The one or more inputs can define, among other things, visual aspects of an information portal and/or an associated or corresponding code, as well as be configured to enable definition of the structure of the information portal and associated content. In some aspects, one or more inputs can define a base structure for an information portal. In some example embodiments, an end user (such as a company or organization) may have a database or repository of content which is not in communication with the online platform, or have specific content for use with an information portal. In some instances, an information portal, and one or more optional sub information portals and a visual code for accessing the information portal can be generated as a template (i.e. information portal template) or shell, where one or more placeholders for content in a standard universal format for content can be included. As an example, in some instances, at least one sub information portal is designated and one or more placeholder links can be generated in the information portal as a link in a standardized format. In some aspects, an information portal (or information portal template, and/or information portal hierarchy information), visual code, and subdomain associated with and directing a device scanning the visual code to the information portal template are simultaneously generated. In some other aspects the information portal template and visual code are sent to an end user (e.g. organization). The information portal can additionally be associated with one more security features for the customization and/or implementation of the information portal, according to other aspects described herein.

In one example embodiment, systems and methods herein can be implemented for on demand generation of an information portal and associated visual code. In some aspects, generation of an on demand information portal can contain multimedia content and/or hyperlinks related to a single or multiple products and sourced from a manufacturers digital resources of product information. Accordingly, in some aspects, a visual code and an information portal and optionally one or more sub information portals can be generated as a template with one or more placeholders in a standardized format, for example placeholders for product registration, operational videos, product manuals, replacement parts and supplies, among others. An end user, such as a manufacturer or distributor can then scan or input a product identifier into the system (e.g. through the online platform, for example by the information portal management component) and the system can automatically search an end user repository, database, or library for content items and/or hyperlinks relating to that product for each of the placeholders in the template. In some instances, a completed template could then be labelled with the product number, assigned a URL, domain, or subdomain in the system, and generate a visual code that can be scanned to access all of the product information. In some aspects, as new information for a product is added into the repository, database, or library or alternatively if information or content is deleted from the repository, the template can be updated either by a user or automatically so that the most current version of relevant information can be viewed in the information portal or sub information portals.

### III. With Reference to the Figures

Referring now to FIG. 1, a flow chart illustrating an example method in the generating and or provisioning of and editing thereof of information related to a visual code. In the example, a code is scanned **102**, wherein the first check is whether the code was provisioned with data in a data repository, checking whether items were already added **104**. Next, in the example, the selection of a private code **106** may be entered and verified **108** to verify access to the materials. This may be useful with regard to medical information or other personal information that a code would be utilized to prevent unauthorized access. The code **106** may also serve to restrict access to users or supplemented with two factor authentication. In the example, if the private code is not verified access will be restricted, otherwise it will display a list of information **112**, such as product information or medical information, or general stored information. Often times, such information is compiled based on like references. For example, a chainsaw product may have a product manual, a user safety video, and a warranty card all within one visual code and stored on a data repository and linked to the visual code.

In some aspects, the information may be editable **114**, and may also have the option to delete **116**. This may be made available through the private access code for the editing of the data stored on the data repository. In other aspects an item may be added, **118**, or made private via a private access code **120**, and or edit or change the private access code **122**. The user then may navigate or click an individual item associated with the information retrieved from the visual code **124**.

Referring now to FIG. 2, an illustration of an example user interface wherein generating a visual code is processed. In the example, a visual code **202** is displayed with a border and a center graphic. The central graphic or logo is one

option upon generation of the visual code that may be specified within the online platform. The option may receive a jpg (jpeg) format or a png format and adapt the logo or graphic for printing central to the visual code. Further, a border option may be placed that allows the generation of a unique border or pattern in which will wrap the exterior of the visual code and provide additional personalization.

Continuing, an example user interface is **204** is disclosed that displays when accessing the visual code **202**. In the example, the visual code points to a location on a database and is displayed as a subdomain or domain. The online platform, such as a WordPress™ platform with plugins and scripts to allow for generation of a visual code from a visual code generator takes user input or input that may be automatically gathered from a rule based or inference engine to generate one or more visual codes that each address to a specific database location as well as a domain name.

Further, in the example of FIG. 2, the file type **208** of item to be stored on the data repository may be specified, allowing for customization of multiple types of files and information, such as a product brochure for a piece of health equipment, along with a video for safe usage, and a warranty card in the event the item needs repairs or warranty information. Additional product information may be sound instructions for the visually impaired, or photos showing use or explaining concepts. The user interface in FIG. 2, for entering information into a data repository, is but one example of how disparate information may be organized and linked to a visual code and a domain name.

Referring now to FIG. 3, an illustration of an example user interface viewing a hyperlink associated with a visual code. In the example, a user may be viewing this interface of curated information showing a quick information portal **300** for a vacation rental which includes information such as a welcome message, house rules, wifi and appliance details, things to do and see in the area, along with emergency contact information. The host may place a visual code that is scannable that will display the curated information on a web browser, thus allowing quick information to be displayed in a curated and visible format. Further, additional domains may be created, such as subdomains for properties. In one example a campus with one property may be assigned one visual code, and the various rooms or housing elements within the campus may have subdomains from the main domain all linked to the same visual code. Thereby allowing rapid deployment of basic or core information, and filtering information more curated to a specific room based on the needs of the room, such as what appliances are available or specific directions from the room.

In the example of FIG. 4, an illustration of an example user interface viewing a hyperlink with information stored in a data repository **400** associated with a visual code. In this aspect, a user is displayed with multiple visual codes **402** linked to various objects and locations, including a user's own history. For example, a user may be interested in a product, by scanning the product with a computing device the product information, such as a quick start video, a user manual, troubleshooting guide, replacement parts, and contact information for the manufacturer may be organized and displayed in a single screen. Thus, overcoming the issue of identifying the product online, searching through pages of information, each of which may be on a different domain. Wherein the present disclosure rapidly adds material from a multitude of domains to a single listing location. As discussed previously, an editing menu allows the creator or a user with an access code to edit the base material, if the contact us page needs updated it can be provisioned through

the generator forms editor option and allow for rapid updating without a need to change the manufacturers website. In this regard manufacturers may see benefits from issuing multiple models of products by allowing updates to product information rapidly and within an easily deployable ecosystem.

Continuing, within the same display or screen a user may view, for example, their flight history by scanning a visual code they printed and hold in their wallet or on their luggage. This allows access to information a user may curate and attach to a visual code, wherein one domain may be for vacations, and subdomains, attached to the same visual code, may show the various locations. Similarly, for flight training schools or for guides for pilots to review the airplane information by scanning a visual code and having a categorized set of information. An additional example is provided for a visual code to provide access to information regarding an outdoor mall or other commercial center, by providing information such as emergency exists, restrooms maps, and security information.

Referring now to FIG. 5, an illustration of an example user interface viewing subdomain information stored on a data repository associated with a visual code. The disclosed illustration shows a hierarchy without the information of the domain and subdomain configuration wherein a single visual code may propagate multiple end URL's, but remain accessible through one scanning image.

Referring now to FIGS. 6-11, wherein illustrations of examples of a user interface of a visual code generator are provided. In the example of FIG. 6, an interface for a visual code generator **600** is disclosed. The code generator accepts user input as well as system provided information to generate a visual code, wherein generating the visual code creates a URL or domain name for the visual code, and provisions a database repository for storage of content. In the example, a type of duration is set with account access to a sample vendor of the quick information portal, wherein different tiered settings may be available. In one aspect the number of codes **602** is the number of visual codes to generate. Additionally, the file name **604** specifies the filename of the visual code and may also be used to provision a database repository file name. The privacy access code **606** grants access to the domain or subdomains. In doing so security is provided for personal identifiable information, such as information like personal health information. In another aspect the open edit allows a customer or end user, as well as an owner or administrator to edit, thereby allowing things like a running maintenance log. Wherein closed edit, or edit turned off allows only the administrator and not the end user the ability to edit information within the database repository or within the generated code.

Continuing, in the example of FIG. 6, the QI options **608** are options for stylizing the visual code. In one aspect a QR Border allows customization with colors for a border. In other aspects it may be lines or other artwork for a border. In another aspect the logo allows for customization of a logo or of graphics or text to be placed within the visual code. Lastly, the editing feature allows a user or anyone with the privacy access code to edit the contents or product information, or information within the data repository. The storage duration **610** allows the visual code to store information for a set period of time, in the example 5 years is specified. However, any duration, including into perpetuity may be set to allow the domain point to a database to remain open for viewing with the scannable visual code.

Referring now to FIG. 7, we see an additional example of a user interface for a visual code generator. Some of the

11

features remain the same from FIG. 6, such as the number of codes to generate **702**, the file name for the codes **704**, and the privacy access code **706**. This example further discloses an option to include an image file on top of the visual code. This option allows a logo or image to be placed central to the visual code. for bringing is disclosed through a check box. Additionally, visual code options **708** allow for setting user editing rights. The storage duration **710** allows a specified duration in which the data repository will hold the information. Lastly, additional options may be customized such as include image feature and enabled hospital notification. In the include image feature an image may be included with the generated code. In the enabled hospital notification, a notification based on hospital codes or other certifications may be generated by the code generator.

Referring now to FIG. 8, an illustration of the generation complete with an alerting function to alert the processing of visual codes. In the example multiple visual codes may be provisioned so as to develop multiple domains and data repository locations so that a plurality of codes may create a plurality of resources, all in which may have underlying database tables. This feature, generating multiple visual codes, is useful with large datasets such as generating visual codes for customer's information or health information, wherein each department of a hospital or each drug, for example, may be associated with a visual code and information provided based on the medical department or drug. In the example, a downloadable comma separated value file allows for additional interaction and processing, such as executing a summary of all created visual codes and information associated therewith using functions that are similar as to excel. Further, the comma separated value file may be incorporated into an analytics module that runs under the platform to share insight, such as scans or access of the visual code. In this regard interoperability is created and may allow for disclosures herein to coordinate with electronic health records or manufacturing records.

Referring now to FIG. 9, yet another example of entering information into the data repository. Wherein the item description **902** is entered which will be used to identify the item. For example, the item description **902** may be User Manual, and the communication **904** may be text, wherein a block of text will then be populated under the visual code and available by scanning the visual code or accessing the generated domain. Additionally, objects can also be made private beneath the privacy access code, as a second layer of protection by using the make code private feature **906**.

FIGS. 10-11, are additional embodiments or examples of entering information into a data repository that will be available by scanning the visual code and accessing a domain name associated with the visual code. In the example of **1002** a text option is selected for uploading a text file. In the example of **1102** we see an audio option selected for uploading audio onto the data repository.

Referring now to FIGS. 12-15, illustrating examples of a user interface for editing the information created and associated with a visual code. In the example of FIG. 12 a user interface is displayed that shows on the left hand side the already created and curated content that may be edited. In the example, a document is selected and allows the uploading of a pdf document to be rendered on a mobile computing device for quick access by a user. The document further allows a user to create a document right in the menu, without having to upload content from a .doc or .docx or pdf. This allows for configuration similar to html for posting documents with text and images directly to the repository for viewing by scanning the visual code.

12

Turning to FIG. 13, an example user interface that may be displayed on WordPress or other content management system, shows the ability to edit or delete items, including also the ability to reorder items or place items within specific subdomains that allows a tiered data approach and the ability to filter down. Returning to the chainsaw example from earlier, the main domain may originate from one visual code that allows subdomains for each model, wherein each model may have their own visual code or may be associated or linked to the main visual code. For example, the has sub-quips option indicates whether or not to create subdomains which originate from the same visual code, but allow accessing under the main domain.

FIGS. 14-15, are additional illustrations showing the editing and removing of an item, as well as selecting a preferential language. Wherein selecting the language will automatically translate the text documents to the language utilizing an API to Google Translate™. Additional elements include making the access from the visual code private, which may be done at creation or post creation, as well as ordering elements. Further, passcodes may be placed on additional files within the data repository for a second or third layer of security. The examples of user interfaces herein are not exhaustive, it is contemplated that the fields may remain the same, however, the interface or design may differ without significantly deviating from the disclosure herein.

Turning now to FIG. 16, a block diagram of an example computing device, such as a mobile computing device. In various embodiments, the computing device may comprise a personal computing device (e.g., smart phone, a computing tablet, a personal computer, laptop, PDA, Bluetooth device, etc.) capable of communicating with the network and scanning a visual code. In the examples herein the computing device is equipped with memory, wherein the memory contains a long-term storage system that is comprised of solid-state drive technology or may also be equipped with other hard drive technologies (including the various types of Parallel Advanced Technology Attachment, Serial ATA, Small Computer System Interface, and SSD). Further, the long-term storage may include both volatile and non-volatile memory components. For example, the processing unit and or engine of the application may access data tables (three dimensional scans) or information in relational databases or in unstructured databases within the long-term storage, such as a SSD. Some other common forms of computer readable media which may also be included herein are, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, CD-ROM, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, RAM, PROM, EEPROM, FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer is adapted to read.

The memory of the example embodiment of a computing device may also contain random access memory (RAM) which holds the program instructions along with a cache for buffering the flow of instructions to the processing unit. The RAM is often comprised of volatile memory but may also comprises nonvolatile memory. RAM is data space that is used temporarily for storing constant and variable values that are used by the computing device during normal program execution by the processing unit. Similar to data RAM, special function registers may also exist, special function registers operate similar to RAM registers allowing for both read and write. Where special function registers differ is that they may be dedicated to control on-chip hardware, outside of the processing unit. Together, the computing module

13

works to execute program instructions, transfer data, generate insights through algorithmic means, and provide intelligence and a way to execute instructions to perform functions herein.

Further, the computing device includes a bus or other communication mechanism for communicating information data, signals, and information between various components of computer device. Continuing, the computing device maintains I/O components through the I/O interface module, such as external devices **1608**, which may be added to improve, enable, or automate functions herein. Wherein a visual code reader may be installed at a site, such as a hospital or a hardware store that allows scanning on products and displaying on a screen at a site location. In this aspect the information is displayed readily at a point of sale or other location that allows accessibility to the user for product or other information. A display may include a touch screen that displays information to user and receives user input from user or a display may be a touch insensitive display. A display may be a mobile device display, a projection display that projects images onto a transparent, partially transparent or opaque structure, or may include any other suitable display technology. The computing device may monitor user input on the touch screen, on any other touch-sensitive device (e.g., a touchpad on a laptop), or using other input components (e.g., a mouse) and may recognize user input for association with transactions, and/or to activate certain functions when scanning and displaying information from the visual code.

Various embodiments of the invention have been described in fulfillment of the various objectives of the invention. It should be recognized that these embodiments are merely illustrative of the principles of the present invention. Numerous modifications and adaptations thereof will be readily apparent to those skilled in the art without departing from the scope of the invention.

The invention claimed is:

1. A computer implemented method for providing rapid access to information on a mobile computing device, comprising:

provisioning an online platform located at a domain name and a database repository supporting the online platform;

generating, by the online platform, at least one visual code that is scannable by the mobile computing device, wherein generating creates a subdomain corresponding to the visual code and provisions a database location within the data repository for each visual code;

generating an information portal and associating it with the at least one visual code directing to the subdomain; gathering web content and associating it with the information portal;

linking one or more pieces of multimedia content to the information portal; and

accessing, by the mobile computing device, the web content and/or multimedia content by scanning the at least one visual code.

2. The method of claim 1, further comprising formatting the web content and/or the multimedia content based on at least the type of content, wherein the formatting identifies it as a photo file, a video file, a pdf file, a sound file, a text file, or a web hyperlink and places a text label associated with a given web content or multimedia content in the information portal.

3. The method of claim 1, further comprising storing the one or more pieces of multimedia content on the database repository of the online platform at the database location.

14

4. The method of claim 3, further comprising updating the one or more pieces of multimedia content on the database repository.

5. The method of claim 1, further comprising displaying, by the mobile computing device, the web content and/or multimedia content in a formatted version that is adapted to the mobile computing device to allow for accessibility and viewing.

6. The method of claim 1, further comprising creating one or more additional subdomains, wherein the creating of additional subdomains allows a reference from the subdomain configured with the at least one visual code.

7. The method of claim 6, further comprising generating one or more sub information portals for each of the one or more additional subdomains, wherein the one or more sub information portals are linked to the information portal.

8. The method of claim 7, further comprising associating web content and/or one or more pieces of multimedia content with the one or more sub information portals.

9. The method of claim 8, further comprising editing, based on an editability field and/or a privacy access code, the web content and/or multimedia content that is associated with the information portal and/or sub information portal, wherein editing changes the web content or the multimedia content associated with the subdomain and/or additional subdomains.

10. The method of claim 1, wherein gathering the web content includes utilizing a web scraper to scrape a domain for the product information.

11. The method of claim 1, wherein generating the at least one visual code that is scannable and also provisioning the database location within the data repository, further comprises the database location generating fields for type of duration, privacy access code, border, logo, editability, and storage duration.

12. The method of claim 11, further comprising selecting, by a user, a privacy access code, wherein the privacy access code provides security for the visual code.

13. The method of claim 11, wherein generating the at least one visual code also generates a logo based on the database location and the field for the logo.

14. The method of claim 11, wherein generating the at least one visual code also generates a border based on the database location and the field for the border.

15. A computer implemented method for providing quick access to information on a mobile computing device, comprising:

provisioning an online platform located at a domain name and a database repository supporting the online platform;

generating at least one visual code that is scannable by the mobile computing device, wherein generating creates a subdomain corresponding to the visual code and provisions a database location within the data repository for each visual code; and

generating an information portal template and associating it with the subdomain such that the visual code directs to the information portal template, wherein the information portal template comprises one or more content placeholders in a standardized format.

16. The method of claim 15 further comprising creating one or more additional subdomains and generating one or more sub information portal templates for each additional subdomain, each sub information portal having at least one content placeholder in a standardized format, wherein each sub information portal template is linked to the information template or another sub information portal.

**15**

**17.** The method of claim **15** further comprising linking at least one of web content and multimedia content to a content placeholder and formatting the content placeholder.

**18.** The method of claim **17**, wherein the linking comprising scanning or inputting a product identifier and automatically searching the web and/or database repository for content to link to each content placeholder. 5

**19.** The method of claim **15**, further comprising generating a privacy access code for the information portal template. 10

**20.** The method of claim **19**, further comprising adding, deleting, modifying web content or multimedia content linked to the information portal template based on the privacy access code.

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15

**16**